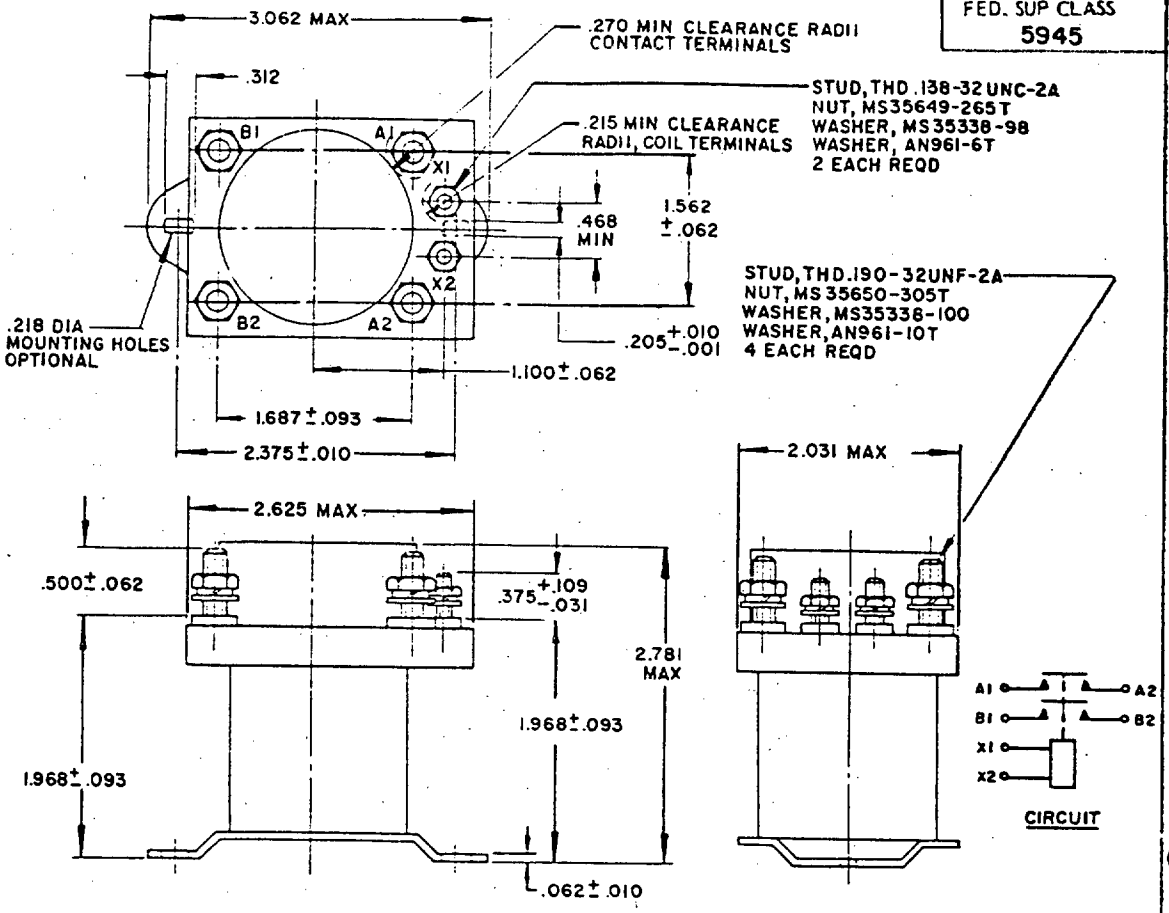
FED. SUP CLASS
5945



INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.001	.03	.138	3.51	.312	7.92	1.968	49.99
.010	.25	.190	4.83	.375	9.53	2.031	51.59
.031	.79	.205	5.21	.468	11.89	2.375	60.33
062	1.57	.215	5.46	.500	12.70	2.625	66.68
.093	2.36	.218	5.54	1.100	27.94	2.781	70.64
.109	2.77	.270	6.86	1.687	42.85	3.062	77.77

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerances are ±.031.
4. Additional flat washer may be used for terminal seat.
5. Part No. MS24178-D1 replaces part No. MS24178-1.
6. For design feature purposes, this standard takes precedence over procurement documents referenced herein.
7. Referenced documents shall be of the issue in effect on date of invitation for bid.

MS PART NUMBER	TYPE	COIL TYPE	TERMINAL TYPE	MOUNTING OR MATING SOCKET	AUXILIARY CONTACTS	MAX WEIGHT IN POUNDS
MS24178-D1	II	DC	STUD	BRACKET	N/A	.75

Ⓖ DENOTES CHANGES

P.A. NAVY-AS
Other Cust USAF 85

International interest

TITLE
RELAY, 55 AMP, 2PST, N.O. TYPE II
NON-HERMETICALLY SEALED

MILITARY STANDARD
MS24178

Procurement Specification
MIL-R-6106

SUPERSEDES:
AN3352

PAGE 1 OF 3

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Project No. 5945-C613-9

APPROVED 20 AUGUST 57 REVISED Ⓕ 6 June 1979 Ⓒ 24 SEPT 82

User activities: Army, Navy, Air Force

Reviewer activities: Army, Navy -AS, EC, Air Force 99, 11

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FED. SUP CLASS 5945

OPERATING CHARACTERISTICS

MS PART NO.	COIL DATA											TIME - MILLISECONDS - MAX 2/					
	Coil	Nominal Volts	Nominal Freq Hz	Ω Res Min	Max Volts	Max Amp	Max Pick-Up Voltage Normal 1/	Max Pick-Up Voltage High temp test	Max Pick-Up Voltage Cont current test	Drop out voltage 1/	Hold voltage 1/	Operate	Release	Bounce Main NO	Bounce Main NC	Bounce Aux NO	Bounce Aux NC
MS24178-D1	X1 X2	28	DC	59	29	.50	18	21	22.5	1.5	7	30	10	3.5	--	--	--

RATED CONTACT LOAD (AMPERES PER POLE) CASE GROUNDED

TYPE OF LOAD	LIFE OPERATING CYCLES X 10 3	28 VDC Main NO	28 VDC Main NC	28 VDC Aux NO	28 VDC Aux NC	115 VAC, 1 PHASE Main 400 Hz	115 VAC, 1 PHASE Main 60 Hz	115 VAC, 1 PHASE Aux 400 Hz	115 VAC, 1 PHASE Aux 60 Hz	115/200 VAC, 3 PHASE 3/ Main 400Hz	115/200 VAC, 3 PHASE 3/ Main 60 Hz	115/200 VAC, 3 PHASE 3/ Aux 400Hz	115/200 VAC, 3 PHASE 3/ Aux 60 Hz	See Appropriate Notes
RESISTIVE	50	55	-	-	-	55	-	-	-	-	-	-	-	
INDUCTIVE	10	40	-	-	-	-	-	-	-	-	-	-	-	
INDUCTIVE														
MOTOR	50	40	-	-	-	35	-	-	-	-	-	-	-	
LAMP														
TRANSFER, LOAD														
MECHANICAL LIFE REDUCED CURRENT	100	14	-	-	-	-	-	-	-	-	-	-	-	
MIN CURRENT	50	5	-	-	-	-	-	-	-	-	-	-	-	

Ⓖ QUALIFICATION BY SIMILARITY

MS PART NUMBER	LOADS TYPE I A	B	C	TYPE I ER D	E	F	DYNAMICS 4/ A	X	Y	ENVIRONMENTAL A	B	C
MS24178-D1	4						2			4		

NOTES, APPLICATION

1/ Over temperature range.
2/ With nominal coil voltage.
3/ Absence of value indicates relay is not rated for 3 phase applications.
Ⓖ 4/ All relays must be tested. Reference MIL-R-6106, Appendix I.

User activities: Army, Navy, Air Force

Reviewer activities: Army, Navy - AS, EC, Air Force - 99, 11

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APPROVED 20 AUGUST 57 REVISED Ⓖ FOR CHANGES SEE PAGES 2 AND 3

P.A NAVY-AS	International interest:	TITLE RELAY, 55 AMP, 2PST, N.O. TYPE II NON-HERMETICALLY SEALED	MILITARY STANDARD MS24178
Other Cust USAF 85			
Procurement Specification MIL-R-6106		SUPERSEDES:	PAGE 2 OF 3

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User activities: Army, Navy, Air Force

Reviewer activities: Army, Navy - AS, EC, Air Force - 99, 11

ENVIRONMENTAL CHARACTERISTICS

TEMPERATURE RANGE . . . . -55° to +71°C

MAX ALTITUDE RATING . . . . . 50,000 FT

SHOCK G-LEVEL . . . . . 25 G

DURATION . . . . . 6-9 ms

MAX DURATION CONTACT OPENING . . 2 ms

VIBRATION - SINUSOIDAL (SEE CHART BELOW)
G-LEVEL . . . . . G

FREQUENCY RANGE . . . . . Hz

VIBRATION-RANDOM
APPLICABLE SPEC . . . . . NA

POWER SPECTRAL DENSITY . . . . . NA

RMS G MIN . . . . . NA

FREQUENCY RANGE . . . . . NA

CURVE . . . . . NA

HIGH SHOCK . . . . . NA

ACCELERATION . . . . . 10 G

ELECTRICAL CHARACTERISTICS

MIN INSULATION RESISTANCE, INITIAL . . . . 100 MEGOHMS
AFTER LIFE OR ENVIRONMENTAL TESTS . . . . 50 MEGOHMS

DIELECTRIC STRENGTH (SEA LEVEL) . . . . . 2-5 SECONDS

	Initial		After Life Tests	
	28VDC	115VAC	28V	115V
COIL TO CASE	1250		1000	
AUX CONTACTS	1250		1000	
ALL OTHER POINTS	1250	1500	1000	1125

DIELECTRIC STRENGTH (ALTITUDE) . . . . . 1 MINUTE

	28VDC	115VAC
COIL TO CASE	500	500
AUX CONTACTS	500	500
ALL OTHER POINTS	500	500

MAX CONTACT DROP INITIAL . . . . . .150 VOLTS
AFTER LIFE TEST . . . . . .175 VOLTS

OVERLOAD CURRENT (NO) . . . . . 440 AMPS

RUPTURE CURRENT (NO) . . . . . 550 AMPS

DUTY RATING . . . . . CONTINUOUS

RFI SPEC MIL-STD-461
(APPLICABLE TO COIL CIRCUITS OF AC OPERATED RELAYS)

DASH NO.	VIBRATION LEVEL				
	5-10 Hz	10-55 Hz	55-250 Hz	250-500 Hz	500-1500 Hz
D1	.08 DA	.06 DA	2 G's	2 G's	

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P.A	International interest	TITLE	MILITARY STANDARD
NAVY-AS Other Cust. USAF 85		RELAY, 55 AMP, 2PST, N.O. TYPE II NON-HERMETICALLY SEALED	MS24178
Procurement Specification MIL-R-6106		SUPERSEDES:	PAGE 3 OF 3

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